



Swathi Sri R 250701769-2025-CSE ▾

S2**Started on** Sunday, 30 August 2026, 7:18 PM**State** Finished**Completed on** Sunday, 30 August 2026, 7:19 PM**Time taken** 1 min 12 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```

23     int complement = x - arr[mid];
24
25     int idx = binary_search(arr, 0, high, complement);
26     if (idx != -1 && idx != mid) {
27         if (arr[mid] < arr[idx]) {
28             *res1 = arr[mid];
29             *res2 = arr[idx];
30         } else {
31             *res1 = arr[idx];
32             *res2 = arr[mid];
33         }
34         return 1;
35     }
36
37     if (find_pair(arr, low, mid - 1, x, res1, res2)) {
38         return 1;
39     }
40     return find_pair(arr, mid + 1, high, x, res1, res2);
41 }
42
43 int main() {
44     int n;
45     if (scanf("%d", &n) != 1) {
46         return 0;
47     }
48
49     int *arr = (int *)malloc(n * sizeof(int));
50     if (arr == NULL) {
51         return 1;
52     }
53
54     for (int i = 0; i < n; i++) {
55         scanf("%d", &arr[i]);
56     }
57
58     int x;
59     if (scanf("%d", &x) != 1) {
60         free(arr);
61         return 0;
62     }
63
64     int res1 = 0, res2 = 0;
65     if (find_pair(arr, 0, n - 1, x, &res1, &res2)) {
66         printf("%d\n%d\n", res1, res2);
67     } else {
68         printf("No\n");
69     }
70
71     free(arr);
72     return 0;
73 }
74

```

	Input	Expected	Got	
✓	4	4	4	✓
	2	10	10	
	4			
	8			
	10			
	14			
✓	5	No	No	✓
	2			
	4			
	6			
	8			
	10			
	100			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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